

Skolem's Theorem in Coherent Logic

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Skolemization in classical FOL

- ▶ Skolemization is the replacement of an axiom of the form

$$\forall \vec{x} \exists y. \phi(\vec{x}, y) \quad (1)$$

by one of the form

$$\forall \vec{x}. \phi(\vec{x}, f_\phi(\vec{x})), \quad (2)$$

where f_ϕ is a fresh function symbol, also called a Skolem function.

- ▶ Skolem's Theorem (conservativity): consequences of (2) not containing f_ϕ already follow from (1).
- ▶ Goal: effective proof transformations (Maehara, 1955), applications in ATP

Skolem's Theorem in constructive FOL

- ▶ Skolem's Theorem fails in constructive FOL=
Consider the following sentence (G. Mints):

$$\forall x_1, x_2 \exists y_1, y_2. P(x_1, y_1) \wedge P(x_2, y_2) \wedge (x_1 = x_2 \rightarrow y_1 = y_2) \quad (3)$$

Clearly, (3) follows from $\forall x P(x, f(x))$. However, (3) does not follow from $\forall x \exists y P(x, y)$. Intuition: you cannot choose the y_i 's correctly without knowing whether the x_i 's are equal or not.

- ▶ Skolem's Theorem holds in constructive FOL (Dowek & Werner)
- ▶ What about coherent logic (CL) as a fragment of FOL= ?

Coherent logic preliminaries

- ▶ Fix a finite first-order signature Σ
- ▶ Coherent implications (sentences): $\forall \vec{x}. (C \rightarrow D)$ with C a conjunction of atoms and D a disjunction of existentially quantified conjunctions of atoms,

$$\forall \vec{x}. (\vec{A} \rightarrow (\exists \vec{y}_1. \vec{B}_1) \vee \cdots \vee (\exists \vec{y}_k. \vec{B}_k))$$

- ▶ Coherent theory: axiomatized by coherent sentences
- ▶ Notation: we leave out the universal prefix, and omit the premiss ' $C \rightarrow$ ' if $C \equiv \top$
- ▶ Discuss: $\exists y. \top$ and $\exists y. \perp$ and $\forall y. \top$ and $\forall y. \perp$
- ▶ Full compliance with Tarski semantics if Σ has a constant or if the theory contains $\exists y. \top$

Examples

- ▶ all usual equality axioms, including congruence
- ▶ $p \vee np$ and $p \wedge np \rightarrow \perp$ (NB $p \vee \neg p$ is **not** coherent)
- ▶ lattice theory: $\exists z. \text{meet}(x, y, z)$
- ▶ geometry: $p(x) \wedge p(y) \rightarrow \exists z. \ell(z) \wedge i(x, z) \wedge i(y, z)$
- ▶ rewriting, $\diamond$ -property: $r(x, y) \wedge r(x, z) \rightarrow \exists u. r(y, u) \wedge r(z, u)$
- ▶ weak- * -elim: $r^*(x, y) \rightarrow (x = y) \vee \exists z. r(x, z) \wedge r^*(z, y)$
- ▶ seriality: $\exists y. s(x, y)$ (who needs a function?)
- ▶ field theory: $(x = 0) \vee \exists y. (x \cdot y = 1)$
- ▶ **not** coherent: Mints' sentence above

History of CL

- ▶ Skolem (1920s): coherent formulations of lattice theory and projective geometry, calling the axioms "Erzeugungsprinzipien" (production rules), anticipating ground forward reasoning. Using CL,
 - ▶ Skolem solved a decision problem in lattice theory
 - ▶ Skolem gave a method to test in/dependence from the axioms of plane projective geometry (example: Desargues' Axiom)
- ▶ Grothendieck (1960s): geometric morphisms preserve geometric logic (= coherent logic + infinitary disjunction). Quite complicated stuff, but we'll stick to Tarski (there is also a forcing semantics of CL).

A proof theory for CL

- ▶ In short: ground forward reasoning with case distinction and introduction of witnesses (ground tableaux reasoning)
- ▶ In full: define inductively $\Gamma \vdash_{\vec{y}}^T A$, where A (Γ) atom (set of atoms) with all variables in $\vec{y}$, in the two cases:
 - (base) A is in Γ , or
 - (step) T has an axiom $\forall \vec{x}. (C \rightarrow (\exists \vec{y}_1.B_1) \vee \dots \vee (\exists \vec{y}_k.B_k))$ such that for some sequence of terms $\vec{t}$ with variables in $\vec{y}$ we have
 - ▶ $C[\vec{t}/\vec{x}]$ is a subset of Γ , and
 - ▶ $\Gamma, B_i[\vec{t}/\vec{x}] \vdash_{\vec{y}, \vec{y}_i}^T A$ for all $i = 1, \dots, n$ (NB $\vec{y}_i$ fresh wrt $\vec{y}$)
- ▶ Rough visualization as a tree with inner nodes like

$$\frac{\Gamma, B_1[\vec{t}/\vec{x}] \quad \dots \quad \Gamma, B_n[\vec{t}/\vec{x}]}{\Gamma} \text{ axiom}$$

- ▶ NB we omit conclusion A in all the nodes, but we should actually keep track of the $\vec{y}, \vec{y}_i$. (In some forcing semantics, pairs like $(\vec{y}; \Gamma)$ are forcing conditions, $\approx$ finite Kripke worlds.)

Derivation trees in CL, example and properties

- ▶ Let T consists of $p \vee \exists x. q(x)$ and $p \rightarrow \perp$ and $q(y) \rightarrow r$
- ▶ Derivation tree for $\emptyset \vdash_{\emptyset}^T r$

$$\frac{\frac{(\perp)}{\{p\}} p \rightarrow \perp \quad \frac{\{q(c), r\}}{\{q(c)\}} q(y) \rightarrow r}{\emptyset} p \vee \exists x. q(x)$$

- ▶ Soundness easily proved by induction on $\Gamma \vdash_{\vec{y}}^T A$
- ▶ NB: $\emptyset \vdash_{\emptyset}^{\forall x. p} p$ not derivable without a constant in Σ
- ▶ So, let's assume a constant in Σ (or $\exists x. \top$, or use $\vdash_{x, \vec{y}}$)
- ▶ Proof of completeness (cf. tableaux, non-constructive):
Develop **fairly** the complete tree of possible derivations, stopping if $\perp$ or A shows up. Infinite branches are models of $\Gamma, \neg A$. (Can be adapted to arbitrary coherent A .)
- ▶ So, classical logic is conservative over CL! (Better: Coste&Coste, Negri)

Skolem constants

Theorem

If T, Γ, A do not mention c and $\Gamma \vdash_{\vec{x}}^{T, P(c)} A$, then $\Gamma \vdash_{\vec{x}}^{T, \exists y. P(y)} A$.

Proof.

If $\Gamma \vdash_{\vec{x}}^{T, P(c)} A$, then $\Gamma, P(c) \vdash_{\vec{x}}^{T, P(c)} A$ by weakening. From the resulting proof we can remove all applications of the axiom $\vdash P(c)$ since $P(c)$ does already occur on the left. We then replace every occurrence of c by a fresh variable u and get a proof of $\Gamma, P(u) \vdash_{\vec{x}, u}^T A$. This substitution operation leaves T, Γ, A unchanged since they do not mention c . It also replaces c by u in instantiations of axioms of T , so that we get a proof in T . Finally, by applying the axiom $\vdash \exists y. P(y)$ we get a proof of $\Gamma \vdash_{\vec{x}}^{T, \exists y. P(y)} A$. □

Decent proof

Assume T, Γ, A do not mention c . Prove each of the following steps by induction on derivation.

$$\Gamma \vdash_{\vec{x}}^{T, P(c)} A \implies (\text{by } \vdash\text{-weakening})$$

$$\Gamma, P(c) \vdash_{\vec{x}}^{T, P(c)} A \implies (\text{still } c \in \Sigma, \text{ but } c \notin T, \Gamma, A)$$

$$\Gamma, P(c) \vdash_{\vec{x}}^T A \implies (u \text{ fresh, } c := u, \text{ now } c \notin \Sigma)$$

$$\Gamma, P(u) \vdash_{\vec{x}, u}^T A \implies (\text{by } T\text{-weakening})$$

$$\Gamma, P(u) \vdash_{\vec{x}, u}^{T, \exists y. P(y)} A \implies (\text{by forward reasoning backwards})$$

$$\Gamma \vdash_{\vec{x}}^{T, \exists y. P(y)} A$$

Beyond Skolem constants, escalation of technicalities

- ▶ We need to replace Skolem terms **by variables**, requiring a new set of 'substitution' lemmas
- ▶ Innermost Skolem terms are important
- ▶ Equality comes with an extra axiom in which the Skolem function occurs, congruence

Possible research directions

1. Why only $\forall x \exists y. P(x, y)$ for atoms?
 - ▶ Must stay coherent, but need not be atom $P(x, y)$
 - ▶ Easy generalization to coherent conclusion format (D)
 - ▶ Unexplored: further generalization to coherent sentences like $A \rightarrow \exists y. P(y)$ and $\forall x. (A(x) \rightarrow \exists y. P(x, y))$
 - ▶ Discuss: non-empty domain, Independence of Premiss, Glivenko class
2. Faster inference (ground inference is slow: Horn counter)
3. Analyze the length of skolemized vs. deskolemized derivations

Metatheoretic results and remarks

- ▶ Corollary of completeness: given a coherent theory T , classically provable coherent sentences are constructively provable
- ▶ For **geometric** logic this is called Barr's Theorem (anticipated by Lawvere and Deligne)
- ▶ Completeness and Barr's Theorem are **not** constructive
- ▶ Barr's Theorem for **coherent** logic can be proved constructively using a cut-elimination argument (Coste & Coste, Negri)
- ▶ Coherent completeness wrt forcing semantics is constructively provable, but does not give the conservativity of classical reasoning